

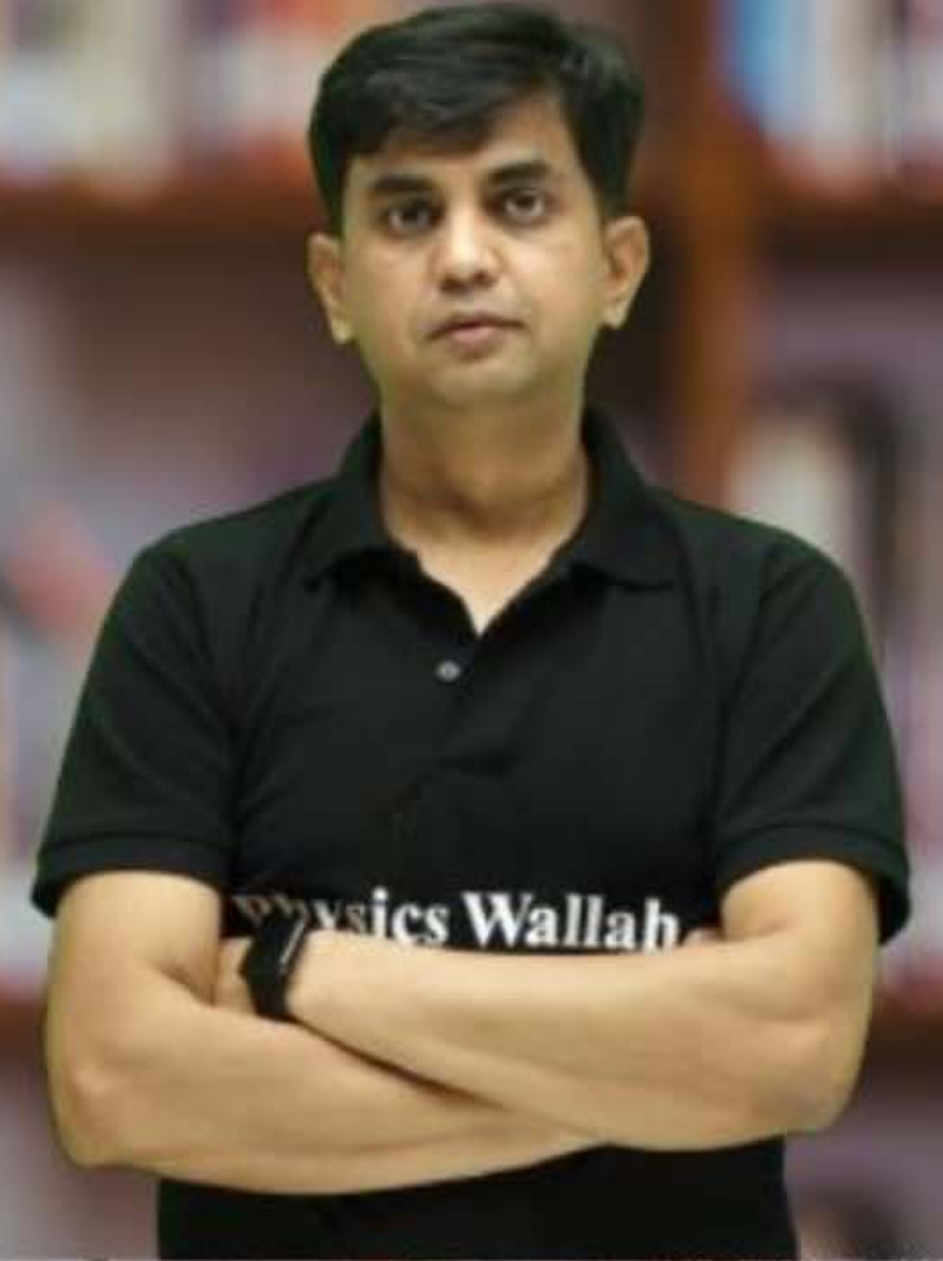
CS & IT ENGINEERING



Computer Network

CIDR Notation

Lecture No. - 04



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Recap of Previous Lecture



Topic

Socket API





Topics to be Covered



Topic

CIDR Notation





Topic : Syllabus



→ CIDR Notation, Network Address Translation;

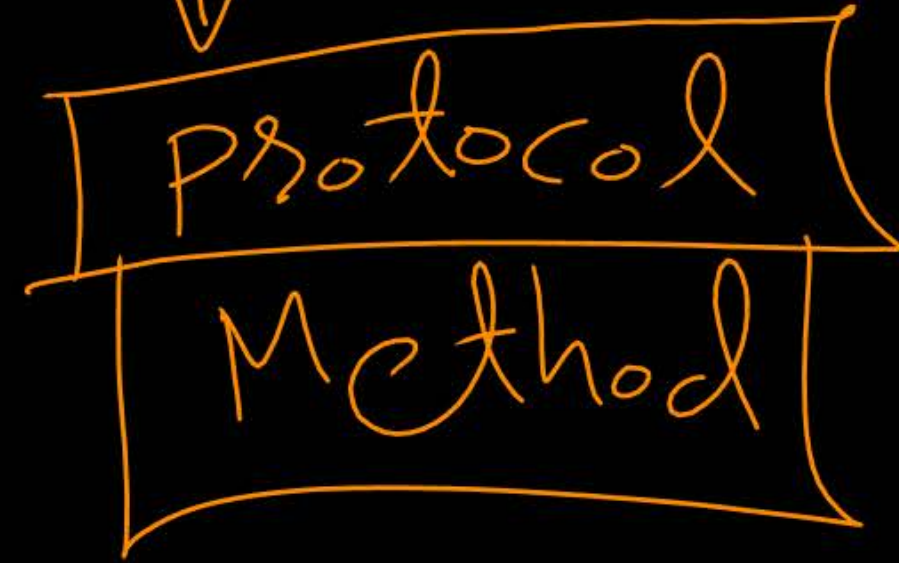
(CS-24)



Topic : CIDR

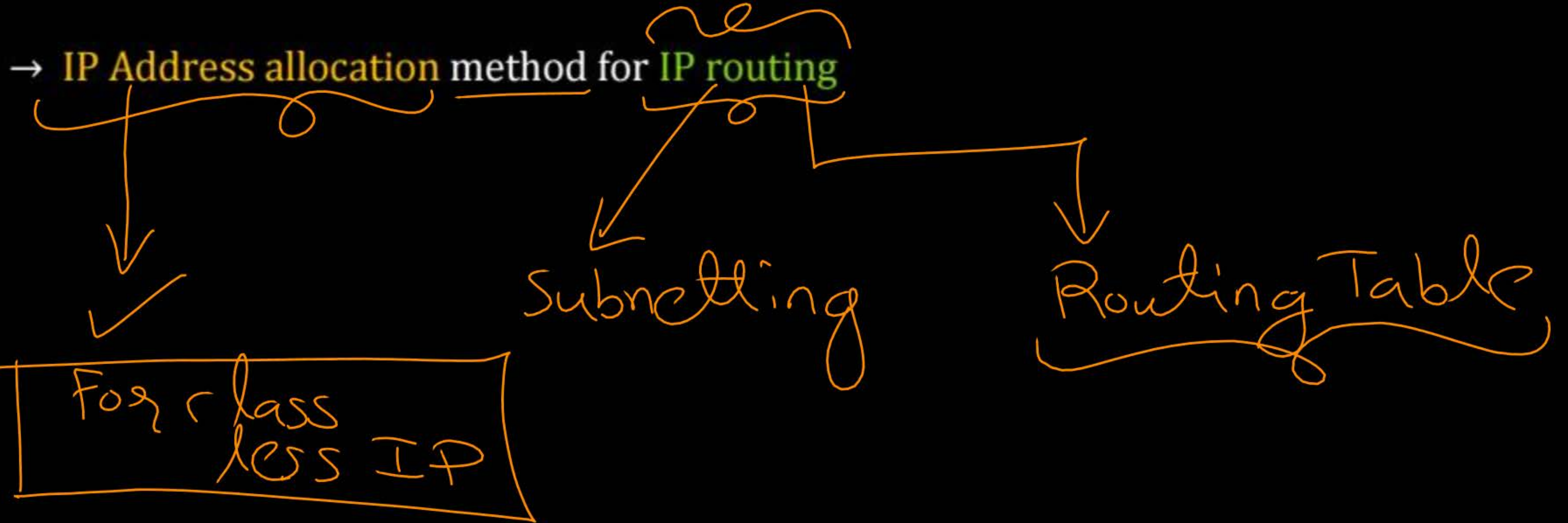


→ CIDR : Classless Inter-Domain Routing





Topic : CIDR





Topic : CIDR



→ Based on Variable-length subnet masking (VLSM)



Topic : CIDR



→ Allows flexibility in creating 'supernets'

↓
For efficient Routing



Topic : IP Address



→ Unique identifier → (Binary string) → Number

→ Assigned to a device that uses Internet Protocol for communication

↓
Host or
Networking Device (Router)



Topic : IP Address



→ IP address have two sections :

1. Network Identifier (Net ID) : x bits
2. Host Identifier (Host ID) : y bits



Topic : IP Address





Topic : IPv4 Address



→ Size of IPv4 Address field = 32 bits (4 bytes)

$$[x + y = 32]$$



Topic : Dotted Decimal Notation

→ 4 Octet, separated by period (.)

→ P . Q . R . S

→ Group of 8



Topic : IPv4 Address



→ IPv4 address are categorised into two type :

1. Classful IPv4 Address

2. Classless IPv4 Address →



Topic : Classless IPv4 Address

→ $P.Q.R.S / x$

→ Prefix / Slash notation

x : number of network id (prefix) bits



Topic : Network Address



→ Used to represent a network

Use :-

⇒ Routing Table



Topic : Network Address



→ Special IP address

→ It can not be assigned to any network interfaces

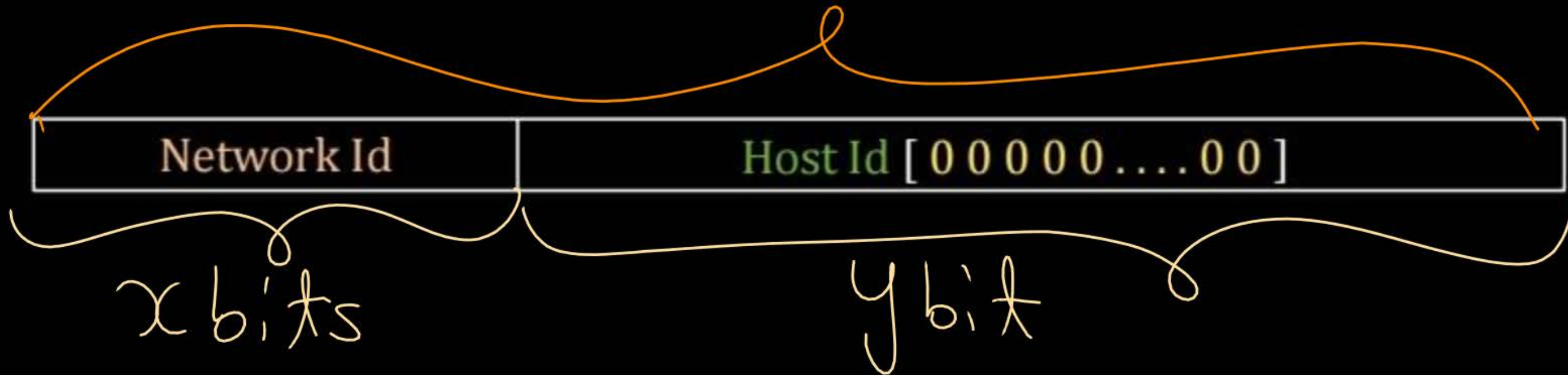


Topic : Network Address



NetID field = Assigned / Given

HostID field = All Zero Bits →





Topic : Broadcast IPv4 Address



- Special IPv4 address (32 bits)
- It can not be assigned to any network interfaces



Topic : Broadcast IPv4 Address

→ Used to **broadcast** a packet (**to all hosts**) of a **network**

* Directed Broadcast Address →

* Limited Broadcast Address
[255.255.255.255]



Topic : Broadcast IPv4 Address

→ Network Directed Broadcast Address

NetID field = Assigned / Given

HostID field = All One Bits



Example 1 :-

[MCQ]



#Q. Consider a class-less IPv4 network, the network address (special IPv4 address) of this network is '165 . 162 . 160 . 0' and size of network identifier field is 20 bits, which one of the following can be a broadcast address (special IPv4 address) of this network ?

- A 165 . 162 . 160 . 255
- ☒ B 165 . 162 . 175 . 255
- C 165 . 162 . 191 . 255
- D 165 . 162 . 255 . 255

Ans: B

Network Address = 165.162.160.0

165.162.1010 0000 00000000

20 bit NetID

12 bit HostID

(All zeros)

Broadcast Address =

165.162.10101111 11111111
 20 bit Net ID 12 bit Host ID

165.162.175.255

All ones



Topic : Network Size



→ Maximum number of IP addresses that can be assigned to a network interfaces



Topic : Network Size



→ Maximum number of hosts can be in a network



Topic : Network Size



→ HostID field (y bits)

Network Size = $[2^y - 2]$ per network

Example 2 :-

[NAT]



#Q. Consider a class-less IPv4 network, the number of bits in the network id field are 20. The maximum number of IP addresses that can be assigned to network interfaces is _____.

Ans = 4096

Number of bits in **Network ID field** (x) = 20 bits

Number of bits in **Host ID field** (y) = $(32 - x) = (32 - 20) = 12$ bits

Number of bits in Host ID field (y) = 12 bits ✓

Maximum number of IP addresses that can be assigned to network interfaces

$$= [2^y - 2] \text{ Addresses}$$

$$= (2^{12} - 2) \text{ Addresses}$$

$$= 4094 \text{ Addresses}$$



Topic : Network Mask



→ Network Mask (**Netmask**) [32-bits]

NetID field = **All One Bits**

HostID field = **All Zero Bits**



Topic : Network Mask

$\Rightarrow$ is not an IP Address
 $\Rightarrow$ it is a binary string



Network Id [1 1 1 1 1 1 1]

Host Id [0 0 0 0 0 0 0]

x bits

y bits

Example 3 :-

[MCQ]



#Q. Consider a class-less IPv4 network where the size of network identifier field is 20 bits. Which one of the following is network mask of this network?

- A 255.255.0.0
- B 255.255.192.0
- ☒ C 255.255.240.0
- D 255.255.255.0

Ans: C

$$\begin{aligned}
 \text{Network Mask} &= \underbrace{11111111 \ 11111111}_{20 \text{ bit Net ID}} \underbrace{11110000}_{12 \text{ bit Host ID}} \underbrace{00000000}_{12 \text{ bit Host ID}} \\
 &= \underline{255} . \underline{255} . \underline{240} . 0
 \end{aligned}$$

[NAT] [GATE-2026, Set-2] [1-Mark]

11T-G



#Q. If an IP network uses a subnet mask of 255.255.240.0, the maximum number of IP addresses that can be assigned to network interfaces is _____.
(answer in integer)

Ans = 4094

Mask = 255 . 255 . 240 . 0

= 11111111 11111111 11110000 00000000

NetID (20 bits)
prefix

12 bit HostID

Number of bits in Host ID field (y) = 12 bits

Maximum number of IP addresses that can be assigned to network interfaces

$$= [2^y - 2]$$

$$= (2^{12} - 2) \text{ Addresses}$$

$$= 4094$$



Topic : Network Mask



IP Address :

Network Id	Host Id
------------	---------

[bit - wise AND] →

Network Mask :

1 1 1 1 1 1 1	0 0 0 0 0 0 0
---------------------	---------------------

Network Address :

Network Id	0 0 0 0 0 0 0
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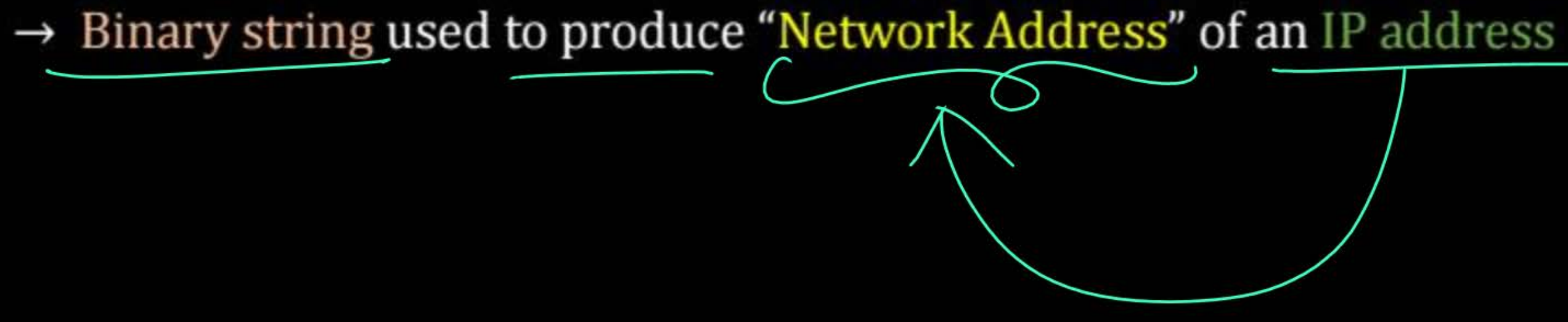
x bits



Topic : Network Mask



→ Binary string used to produce “Network Address” of an IP address





Topic : Network Mask



Network Address ← IP Address [bit-wise AND] Netmask



Topic : Network Mask



→ All hosts belongs to same network must have same "Network Address"

[MCQ] [GATE-2010] [2-Mark] IIT-G, H.W.

#Q. Suppose computers A and B have IP addresses "10 . 105 . 1 . 113" and "10 . 105 . 1 . 91" respectively and they both use the same netmask N. Which of the values of N given below should not be used if A and B should belong to the same network?

- A 255 . 255 . 255 . 0
- B 255 . 255 . 255 . 128
- C 255 . 255 . 255 . 192
- D 255 . 255 . 255 . 224



2 mins Summary



Topic

CIDR Notation



THANK - YOU